

CLUSTERING

CLUSTER VALIDITY – PART IV

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CLUSTER VALIDITY

The following concepts will be introduced:

✓ **RELATIVE INDICES**

- KALINSKI AND HARABASZ
- DUNN
- DAVIES-BOULDIN
- AKAIKE INFORMATION CRITERION
- BAYESIAN INFORMATION CRITERION
- MINIMUM DESCRIPTION LENGTH

✓ **FUNDAMENTAL PROBLEM OF CLUSTER VALIDITY**

INTERNAL AND EXTERNAL INDICES require statistical testing, which could become **COMPUTATIONALLY INTENSIVE**.

RELATIVE CRITERIA eliminate such requirements and concentrate on the comparison of clustering results generated by different clustering algorithms or the same algorithm with different input parameters.

THE FUNDAMENTAL PROBLEM OF CLUSTER VALIDITY

Determine the "**TRUE NUMBER OF CLUSTERS K** ".



Project the data into a 2 or 3 dimensional Euclidean space and to use visualization techniques which can provide some insights on the number of clusters.

Far from being sufficient to represent the true data structures, therefore, the visual estimation of K can only be restricted to a very small scope of applications.

The **MAIN INDICES** are

➤ **CALINSKI AND HARABASZ**

- The value of K corresponding to the maximum is taken to be the optimal number of clusters

➤ **DUNN**

- The value of K corresponding to the maximum is taken to be the optimal number of clusters (a large value suggests the presence of compact and well-separated clusters)

➤ **DAVIES-BOULDIN**

- The value of K corresponding to the minimum is taken to be the optimal number of clusters

For **PROBABILISTIC MIXTURE MODEL-BASED CLUSTERING** the **MAIN INDICES** are

➤ **AKAIKE INFORMATION CRITERION (AIC)**

- The value of K corresponding to the minimum is taken to be the optimal number of clusters

➤ **MINIMUM DESCRIPTION LENGTH (MDL)**

- The value of K corresponding to the minimum is taken to be the optimal number of clusters

➤ **BAYESIAN INFORMATION CRITERION (BIC)**

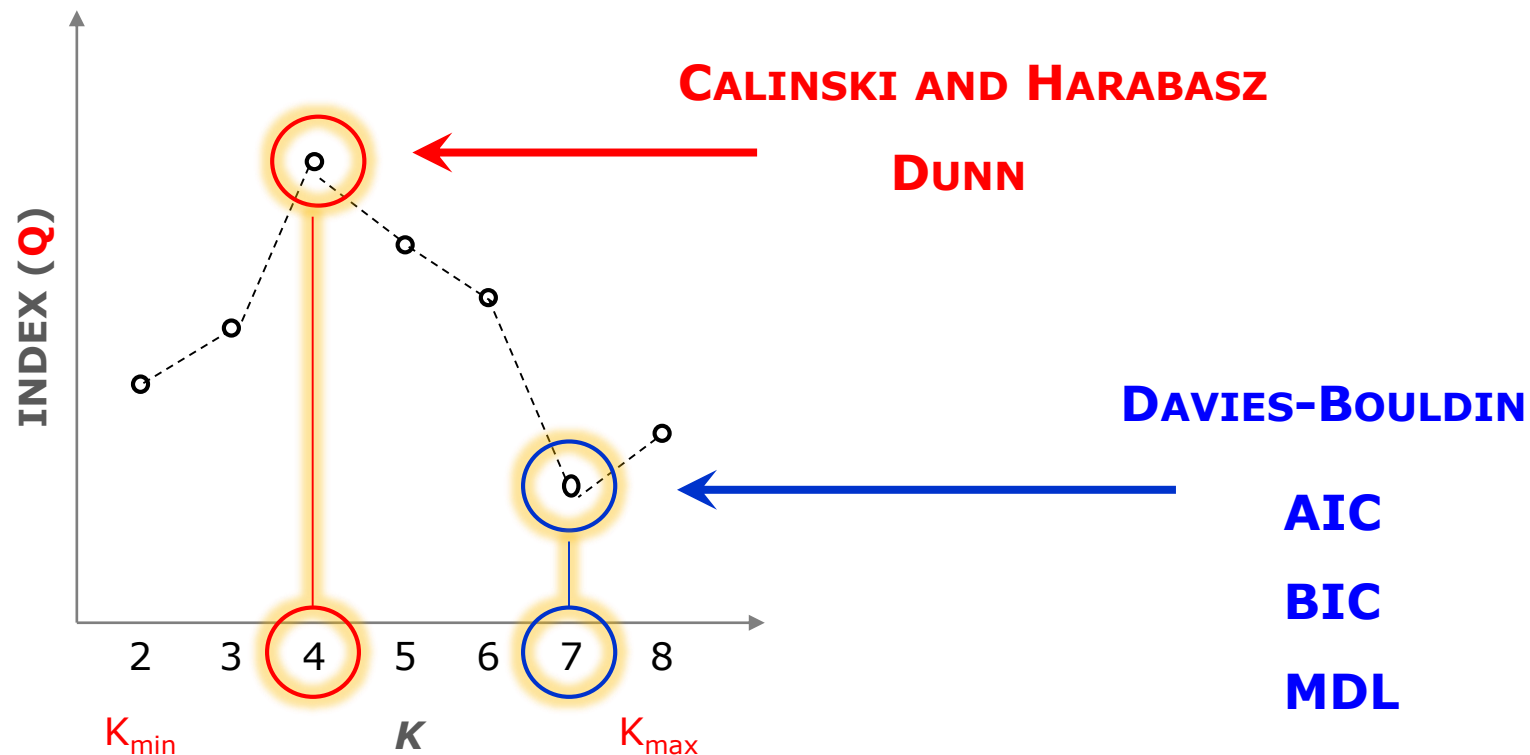
- The value of K corresponding to the minimum is taken to be the optimal number of clusters

For a clustering algorithm that requires the input of the number of clusters K from users, a sequence of clustering structures is obtained by running a clustering algorithm several times (r) where K ranges from $K_{\min}$ to $K_{\max}$.

GENERATE SEQUENCE OF CLUSTERING STRUCTURES

1. Choose a clustering algorithm and a validity index
2. **FOR** $K=K_{\min}$ to $K_{\max}$ **DO**
3. **FOR** $i=1$ to r **DO**
4. run the clustering algorithm with K and use parameter values different from the previous running
5. compute the value q of the validity index and set $q(i) = q$
6. **END FOR**
7. Choose the best value q^* of the validity index $\{q_1, \dots, q_r\}$
8. set $Q(K) = q^*$
9. **END FOR**

The clustering structures are then evaluated based on the computed index, and the “optimal clustering solution” is determined by choosing the one with the best value of the index.



In the case of hierarchical clustering structures, the indices are also known as stopping rules, which tell where the best level is in order to cut the dendrogram.